

A Multi-Source Dataset on Constituency Development Fund Projects and Municipal Financial Activities of Lusaka City Council, Zambia

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Abstract

This data article describes a multi-source dataset compiled from publicly available digital records associated with Lusaka City Council (LCC), Zambia, the capital and one of the country's largest local authorities. The dataset was developed as part of a CSC 4792 (Data Mining and Warehousing) practical assignment requiring the extraction, cleaning and curation of information relating to local-government financial and development activities. The current curated release contains 593 structured records distributed across seven thematic sub-datasets: CDF projects (301 records), CDF allocations (19 records), budgets (31 records), expenditure (28 records), financial reports (96 records), Local Government Equalisation Fund (LGEF) records (55 records) and revenue (63 records). The CDF project records cover all seven Lusaka constituencies (Kanyama, Matero, Munali, Mandevu, Lusaka Central, Kabwata and Chawama) and are classified into nine project categories, with Roads and Drainage and Education accounting for the largest numbers of records. Data were collected through web scraping of council digital sources, PDF and OCR extraction from constituency schedules and financial statements, and manual verification against official source documents. Cleaning included duplicate checking, removal of extraction artefacts, standardisation of project and status fields, and structural validation. The dataset supports analysis of local-government development-project implementation, CDF allocation and utilisation, municipal budgeting, revenue mobilisation and financial performance in Zambian local authorities.

Keywords: Lusaka City Council; Local Government Act; Constituency Development Fund; Zambia; municipal finance; budget estimates; local revenue; Local Government Equalisation Fund; data mining; Copperbelt-style local governance data

Specifications Table

Subject	Value
Subject	Social Sciences — Public Administration and Local Governance
Specific subject area	Municipal finance, local government budgeting, Constituency Development Fund (CDF) project implementation, CDF allocation and utilisation
Type of data	Structured tabular data (CSV files)
Data format	Pipe-separated values (CSV with delimiter)
How data was acquired	Web scraping of Lusaka City Council digital sources; programmatic PDF/OCR extraction (constituency schedules, financial statements, budget documents); manual verification and curation against official source documents
Data source location	Lusaka City Council, Lusaka, Zambia
Data accessibility	Repository: Kaggle (https://www.kaggle.com/datasets/peterkay86/lusaka-city-council-cdf-and-financial-data); Code: GitHub (https://github.com/PeterKay86/unza-csc4792-lusaka-city-council-dataset.git)
Related research article	This is a data description article; no associated research article at time of publication

1 Value of the Data

1. Local government financial transparency. The dataset provides a structured, publicly reusable extraction of Lusaka City Council's budgets, expenditure, revenue, financial reports and LGEF records covering 2020–2023, enabling longitudinal analysis of municipal financial performance in Zambia's capital.
2. CDF implementation tracking. With 301 project records spanning all seven Lusaka constituencies and nine project categories, the dataset offers granular visibility into how Constituency Development Fund allocations are planned, approved and deferred.
3. CDF allocation and utilisation analysis. The 19-record CDF allocation table, covering 2023–2026, tracks approved allocations, disbursements, amounts spent and unspent balances at both constituency and city-wide level.
4. LGEF utilisation analysis. The 55-record LGEF sub-dataset supports analysis of intergovernmental fiscal transfer effectiveness for Lusaka City Council.
5. Provenance and reproducibility. The CDF component retains source documents, extracted text, OCR outputs and a source manifest, allowing individual records to be traced back to the original constituency schedule, financial statement or budget document.
6. Reuse potential. Researchers studying Zambian local governance, comparative municipal finance, CDF effectiveness, or data mining and information extraction from semi-structured government documents can reuse this dataset; it is also suitable for teaching data cleaning, ETL and data-warehousing workflows.

2 Data Description

2.1 Dataset Overview

The dataset consists of 593 structured records distributed across seven thematic CSV files, each using the pipe (|) character as the field separator. Final files follow the naming convention db-unza26-csc4792-lusaka_city_council_[DESCRIPTION].csv.

File (thematic content)	Rows	Fields	Years covered
cdf_projects	301	20	2024–2025
cdf_allocations	19	12	2023–2026
budgets	31	13	2020–2023
expenditure	28	—	2020–2023
financial_reports	96	—	2020–2023
lgef	55	—	2020–2023
revenue	63	—	2020–2023

The seven datasets are maintained as separate but related tables rather than being forced into a single flat table, preserving the different meanings and granularities of project, allocation and financial records.

2.2 Column Schema

Unlike a single unified schema, the CDF and financial sub-datasets retain schemas appropriate to their record type.

CDF project dataset (20 fields)

- record_id
- financial_year
- constituency
- ward
- project_name
- project_category
- project_description
- approved_budget_zmw
- amount_disbursed_zmw
- amount_spent_zmw
- project_status
- completion_percentage
- approval_date
- completion_date
- beneficiaries
- contractor
- source_title
- source_url
- date_accessed
- notes

CDF allocation dataset (12 fields, principal fields)

- financial_year
- constituency
- allocation_category
- approved_allocation
- amount_disbursed
- amount_spent
- unspent_balance
- source information
- explanatory notes

Budget dataset (13 fields)

- year
- entity
- transaction_type
- line_item
- original_budget
- adjustments
- final_budget
- actual_amount

- performance_percentage
- variance_zmw
- variance_percentage
- currency
- source_document

The expenditure, financial-reports, revenue and LGEF datasets record year, category/section, line item, amount (ZMW) and the originating source document, allowing each value to be traced to the relevant financial statement.

2.3 Record Types

CDF projects are classified by constituency, project category and project status.

Constituency	Projects
Kanyama	54
Matero	51
Munali	47
Mandevu	46
Lusaka Central	42
Kabwata	38
Chawama	23
Total	301

Project category	Records
Roads and drainage	127
Education	72
Health	33
Other community infrastructure	20
Public safety	14
Markets and commerce	12
Equipment and vehicles	9
Community and recreation	8
Water and sanitation	6

Project status	Records
Planned	110
Deferred	112
Approved	79
Total	301

Project-status values were not inferred from missing information; they were derived from the underlying source schedules. The 112 Deferred records originate from the deferred-project schedule, which indicated that those projects had not been approved because of insufficient funding.

The financial sub-datasets (budgets, expenditure, financial reports, revenue, LGEF) record council-wide financial activity by year, category and line item rather than by constituency.

2.4 Fiscal Year Coverage

Sub-dataset	Records	Years	Key sources
CDF projects	301	2024–2025	2025 community-project schedule; 2024 constituency documents; deferred-project schedule
CDF allocations	19	2023–2026	CDF allocation and financial records
Budgets	31	2020–2023	LCC financial statements and budget documents
Expenditure	28	2020–2023	LCC financial statements
Financial reports	96	2020–2023	LCC financial statements
Revenue	63	2020–2023	LCC financial statements
LGEF	55	2020–2023	LCC financial statements

Source documents for the financial component span 2018–2023, but the current structured tables provide reliable coverage only for 2020–2023; the presence of a source document does not imply that every year contains all fields required for the final schema.

3 Methods

3.1 Data Acquisition

The primary institutional source was the Lusaka City Council official website, identified in the project specification as the assigned council source for Project Team 1. For the CDF component, the repository retains 34 scraped CDF-related articles, 34 initial project-candidate records, an 18-record source manifest, 18 retained source documents, extracted text, OCR text, extracted tables, and source URLs with access dates. Retained source documents include constituency-specific CDF documents, LCC community-project schedules, budget documents, financial statements and a deferred-project schedule. The financial component was constructed from LCC financial statements covering 2018–2023 retained in the raw-data collection.

3.2 Data Extraction

CDF-related web content was collected through a dedicated scraping script, generating an article dataset of 34 records from which candidate project records were derived. Several CDF source documents were PDFs, including scanned constituency schedules, so the repository retains both extracted text and OCR text; where table structure was sufficiently recoverable, project attributes such as project name, constituency, ward and approved amount were extracted programmatically. OCR-derived information was subsequently reviewed because OCR can alter names, numbers and table relationships. Automated extraction was not treated as final authority, since web pages and documents contain navigation elements, repeated content and inconsistent layouts.

3.3 Data Cleaning

The cleaning process preserved the original raw data while producing separate cleaned datasets. The principal cleaning operations were:

1. Removal of duplicate or repeated records where appropriate.
2. Removal of extraction artefacts.
3. Standardisation of column names.
4. Standardisation of project-status values.
5. Standardisation of constituency values.
6. Cleaning and preservation of ward values (left blank rather than inferred where not confidently recoverable).
7. Conversion of financial fields to numeric representations.
8. Retention of source information for traceability.
9. Explicit handling of missing values, distinguishing "not reported" from zero.
10. Structural validation of the final datasets.

3.4 Data Integration

The seven sub-datasets were kept as separate, related tables rather than merged into a single flat file, in order to preserve the different meanings and granularities of project, allocation and financial records. Project verification combined automated candidate extraction with direct inspection of official source documents; the resulting CDF quality report records 301 project records, 20 project columns and 19 allocation records.

3.5 Deduplication

Duplicate checking was applied during cleaning (Section 3.3, step 1). The final CDF quality report records zero duplicate project keys and zero projects missing source information across the 301 project records.

3.6 Handling Source Discrepancies

Source totals were not silently altered to force agreement with calculated totals. For the 2025 Kabwata projects, the amounts sum to ZMW 16,789,272 against a printed source total of ZMW 16,789,271 (a ZMW 1 difference). For Kanyama, the amounts sum to ZMW 17,001,860 against a printed total of ZMW 17,001,859 (ZMW 1 difference). For Munali, the amounts sum to ZMW 17,656,230 against a printed total of ZMW 17,656,232 (ZMW 2 difference). A further discrepancy occurs in the 2024 CDF subprogramme information: the relevant table reports community projects of ZMW 122,236,211 against a total CDF allocation of ZMW 214,449,494, while a nearby narrative refers to approximately ZMW 128.3 million for community projects; the exact subprogramme-table value was retained. These discrepancies were documented rather than corrected without evidence.

3.7 Thematic Organisation and File Naming

The assignment specifies that dataset files must be in CSV format, use the pipe character (|) as the field separator, and follow the naming convention db-unza26-csc4792-[DESCRIPTION].csv. The CDF final datasets already follow this convention; the financial datasets are renamed to the same convention for final submission:

- db-unza26-csc4792-lusaka_city_council_cdf_projects.csv
- db-unza26-csc4792-lusaka_city_council_cdf_allocations.csv

- db-unza26-csc4792-lusaka_city_council_budgets.csv
- db-unza26-csc4792-lusaka_city_council_expenditure.csv
- db-unza26-csc4792-lusaka_city_council_financial_reports.csv
- db-unza26-csc4792-lusaka_city_council_lgef.csv
- db-unza26-csc4792-lusaka_city_council_revenue.csv

4 Limitations

1. Coverage scope: the dataset represents publicly available digital records rather than the complete internal administrative database of Lusaka City Council.
2. Source quality: source documents differ considerably in structure and quality; some constituency records were available only as scanned documents requiring OCR, which can introduce errors in project names, amounts and table relationships.
3. Incomplete project attributes: the CDF project table does not provide disbursement, expenditure, completion percentage, completion date, beneficiary or contractor information for the current 301 records; approved budgets are absent for 191 records and approval dates for 222 records.
4. Ward completeness: some ward information could not be recovered confidently from scanned documents and was deliberately left blank rather than guessed.
5. Temporal coverage of financial tables: the financial tables provide structured coverage for 2020–2023 even though raw financial-statement documents covering earlier years (from 2018) are retained in the repository.
6. Domain completeness: the current release does not yet contain dedicated final tables for every administrative domain identified in the assignment specification, particularly Integrated Development Plans, council programmes, committees, wards and council resolutions/reports.

5 Ethical Considerations

All source documents used in this dataset are publicly available from the Lusaka City Council website and related council digital sources. No confidential, personal or restricted-access data was used. CDF project records identify constituencies, wards, project names and contractors in their official administrative capacity, which are matters of public record; the beneficiaries field is currently unreported across all 301 project records, so no personal data of private individuals is included in the released dataset. This work was completed as part of a university course assignment (CSC 4792, University of Zambia) using only publicly accessible council information.

CRedit Author Statement

Peter K. Lubasi: Project administration, data integration, schema design, data curation, data validation, repository management and manuscript preparation.

Vanessa: CDF research, source identification, CDF allocation and project data collection and verification.

Blessings: Financial research, financial-document extraction and preparation of budget, LGEF, revenue, expenditure and financial-report datasets.

Gorette: Development-project and council-administration research, including development projects, IDPs, council programmes, wards, committees and resolutions/reports.

Nicholas: Data cleaning, duplicate checking, missing-value analysis, standardisation, validation, metadata and documentation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

The dataset is publicly available on Kaggle: <https://www.kaggle.com/datasets/peterkay86/lusaka-city-council-cdf-and-financial-data>

The source code for data extraction, cleaning, and integration is available on GitHub: <https://github.com/PeterKay86/unza-csc4792-lusaka-city-council-dataset.git>

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